How the two branches see the EEG differently

	LinOSS branch (OscSeq)	CatBoost branch
Input montage	**19 referential electrodes** one 19 × 100 token per second (TUSZ)	**16 longitudinal bipolar chains** formed from the 19 referential electrodes
Frequency content	**log|FFT| of each 1 s, bins 0–99 Hz** unfiltered, z-scored	**0.5–35 Hz band-pass + 60 Hz notch** zero-phase filters, resampled to 200 Hz
Time resolution	**1 s tokens** one logit per second	**60 s windows, 15 s stride** 588 features per window
Temporal context	**learned oscillators over 240 s** natural periods initialised 5–240 s windows every 120 s, sin² blending	**±7-window summary statistics** 11 statistics per feature, ±105 s p_C(t) reads at most 165 s either side of t

Measured behaviour on the TUSZ eval set

Threshold crossing vs. seizure (median)	**crosses its threshold 16 s after onset** and drops below it 14 s before the end	**crosses its threshold 24.5 s before onset** and stays above it 18.5 s after the end
False-alarm seconds within 60 s of a seizure	**3.5 %** of the LinOSS-only false-alarm seconds	**59.9 %** of the CatBoost-only false-alarm seconds
Logit correlation of the two branches	**Pearson r = 0.50 on background seconds and 0.48 on ictal seconds**	